

Yulia Alexandr

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Research interests	algebraic statistics, applied algebraic geometry, mathematical machine learning	
Employment	University of California, Los Angeles	Jan 2024–Jun 2026
	Hedrick Assistant Adjunct Professor	
	Mentor: Guido Montúfar	
	Harvard University	Jun–Sep 2025
Education	Postdoctoral Fellow in Applied Mathematics	
	Mentor: Anna Seigal	
	University of California, Berkeley	2019–2023
	PhD in Mathematics	
Awards and fellowships	Advisors: Bernd Sturmfels and Serkan Hoşten	
	Thesis: <i>From Voronoi Cells to Algebraic Statistics</i>	
	Wesleyan University	Class of 2019
	BA in Mathematics with High Honors; class rank: 1	
Awards and fellowships	Advisor: Karen Collins	
	Thesis: <i>Combinatorial Nullstellensatz: Various Proofs, Extensions & Applications</i>	
	DARPA AIQ grant (PI: Guido Montúfar)	2025-2026
	<i>key personnel, contributed to proposal writing and project development</i>	
	AMS-Simons travel grant (\$5000)	2025-2026
	AWM travel grant (\$2300)	2024
	UC Berkeley conference travel grant (\$900)	2022
	NSF Graduate Research Fellowship	2020-2023
	Chancellor's Graduate Fellowship (UC Berkeley)	2019-2020
	Phi Beta Kappa (Connecticut Gamma Chapter)	2019
	Rice Prize (Wesleyan University)	2019
	<i>awarded to a senior for excellence in mathematics</i>	
	Rae Shortt Prize (Wesleyan University)	2018
	<i>awarded to a junior for excellence in mathematics</i>	
	Twin Cities REU at the University of Minnesota	2018
	DIMACS REU at Rutgers University and Charles University (Prague)	2017
	Treespace REU at Lehman College	2016

Stress-testing neural network verifiers with provably robust instances

with David Troxell, Sofia Hunt, Stephanie Lei, and Guido Montúfar.

Submitted, available on [arXiv](#), 2026.

Algebraic invariants of lightning self-attention

with Hao Duan and Guido Montúfar.

Submitted, available on [arXiv](#), 2026.

Robustness verification of polynomial neural networks

with Hao Duan and Guido Montúfar.

Submitted, available on [arXiv](#), 2026.

Decomposing determinantal varieties from statistics via matroid theory

with Per Alexandersson, Emiliano Liwski, Fatemeh Mohammadi, and Pardis Semnani.

Submitted, available on [arXiv](#), 2026.

Constraining the outputs of ReLU neural networks

with Guido Montúfar.

To appear in *Theoretical Foundations of Deep Learning*, 2025.

Decomposing conditional independence ideals with hidden variables

with Kristen Dawson, Hannah Friedman, Fatemeh Mohammadi, Pardis Semnani, and Teresa Yu.

Published in *Advances in Applied Mathematics*, **174** (2026), paper no. 103003.

New directions in algebraic statistics: three challenges from 2023

with M. Bakenhus, M. Curiel, S. K. Deshpande, E. Gross, Y. Gu, M. Hill, J. Johnson, B. Kagy, V. Karwa, J. Li, H. Lyu, S. Petrović, and J. I. Rodriguez.

Published in *Algebraic Statistics*, **15** (2024), no. 2, 357–382.

Mixtures of discrete decomposable graphical models

with Jane Ivy Coons and Nils Sturma.

Published in *Algebraic Statistics*, **15** (2024), no. 2, 269–293.

Maximum information divergence from linear and toric models

with Serkan Hoşten.

Published in *Information Geometry*, **8** (2025), 159–197.

Moment varieties for mixtures of products

with Joe Kileel and Bernd Sturmfels.

Published in Proceedings of the *International Symposium on Symbolic and Algebraic Computation (ISSAC 2023)*, ACM, New York, 2023, 53–60.

Decomposable context-specific models

with Eliana Duarte and Julian Vill.

Published in *SIAM Journal on Applied Algebra and Geometry* **8** (2024), no. 2, 363-393.

Logarithmic Voronoi cells for Gaussian models

with Serkan Hoşten.

Published in *Journal of Symbolic Computation* **122** (2024), paper no. 102256.

Logarithmic Voronoi polytopes for discrete linear models

Published in *Algebraic Statistics* **15** (2024) no. 1, 1–13.

Logarithmic Voronoi cells

with Alexander Heaton.

Published in *Algebraic Statistics* **12** (2021), no. 1, 75-95.

Computing a logarithmic Voronoi cell

with Alexander Heaton and Sascha Timme.

Published online at [HomotopyContinuation.jl](#), 2019.

Recovering conductances of resistor networks in a punctured disk

with Brian Burks, Sunita Chepuri, and Patricia Commins.

Preprint. Available on [arXiv](#), 2019.

Deformations of the Weyl character formula for $SO(2n + 1, \mathbb{C})$ via ice models

with P. Commins, A. Embry, S. Frank, Y. Li, and A. Vetter.

Preprint. Available on [arXiv](#), 2018.

Growth of meandric numbers

with Kayla Cummings and Edgar Jaramillo Rodriguez.

Transcription in *DIMACS-DIMATIA REU booklet* (pp. 33-36), 2017.

Teaching

Instructor (UCLA)

PIC 10B: Intermediate Programming (C++) Winter, Spring 2025, Winter 2026

PIC 10A: Introduction to Programming (C++) Winter, Fall 2024

PIC 16A: Python with Applications I Spring 2024

Graduate student instructor (UC Berkeley)

MATH 10B: Methods of mathematics Spring 2023

MATH 54: Linear algebra and differential equations Fall 2022

Directed reading program mentor (UC Berkeley)

Topics: algebraic combinatorics and graph theory Spring 2020

Teaching assistant (Wesleyan University)

MATH 231: Probability theory

Fall 2017, Fall 2018

MATH 274: Graph theory

Spring 2018

Talks

key: ★=invited/colloquium; †=contributed; Δ=seminar/lecture.

Algebraic invariants in machine learning

★ Meeting on Applied Algebraic Geometry (MAAG) at Clemson April 2026

Δ Boston College Math & Machine Learning seminar April 2026

Algebraic methods in statistics and machine learning

★ Naval Postgraduate School operations research colloquium October 2024

★ Vanderbilt University mathematics colloquium November 2025

★ Wesleyan University mathematics and CS colloquium November 2025

★ University at Albany mathematics colloquium December 2025

★ Colby College mathematics colloquium December 2025

★ The University of Melbourne mathematics colloquium January 2026

★ University of Florida mathematics colloquium January 2026

★ UMass Boston mathematics colloquium February 2026

Constraining the outputs of ReLU neural networks

★ New Directions in Algebraic Statistics at IMSI, Chicago July 2025

★ AMS special session on algebra and discrete mathematics in machine learning
at Cal Poly, SLO May 2025

Can algebra be applied?

★ Worcester Polytechnic Institute (WPI) math colloquium August 2025

★ UMSA Professor talk at UCLA March 2025

Mixtures of discrete decomposable graphical models

★ SIAM Conference on Applied Algebraic Geometry in Madison July 2025

★ AMS special session on discrete and algebraic methods in mathematical biology at Cal Poly, SLO May 2025

Δ UW-Madison Applied Algebra Seminar November 2024

Maximum information divergence from linear and toric models

Δ Math Machine Learning seminar MPI MIS + UCLA February 2024

Δ Caltech Information, Geometry & Physics seminar May 2026

Moment varieties for mixtures of products

† International Symposium on Symbolic and Algebraic Computation (ISSAC)
in Tromsø July 2023

- ★ AMS special session on mathematics in data science at Spring Western sectional meeting May 2023
- ★ SFSU Algebra, Geometry, and Combinatorics day March 2023

Computing logarithmic Voronoi cells

- ★ AMS special session on polynomial systems, homotopy continuation and applications at the JMM January 2023

Decomposable context-specific models

- ★ CEG Workshop in Warwick (virtual) September 2022

Logarithmic Voronoi cells

- ★ Naval Postgraduate School in Monterey CA June 2023
- ★ Santa Clara University math and CS colloquium October 2022
- △ Discrete Math & Geometry seminar at TU-Berlin (virtual) October 2022
- △ SFSU Algebra, Geometry, and Combinatorics Seminar October 2021
- △ Berkeley Combinatorics Research Seminar (virtual) April 2021
- △ Nonlinear Algebra Seminar Online (virtual) April 2020

Logarithmic Voronoi polytopes

- △ Mathematical Methods in Data Analysis in Tirana, Albania July 2022

Combinatorics of logarithmic Voronoi cells

- △ Algebra and Geometry seminar at University of Magdeburg July 2022

Logarithmic Voronoi cells for Gaussian models

- ★ SIAM Conference on Applied Algebraic Geometry in Eindhoven July 2023
- † Effective Methods in Algebraic Geometry in Kraków, Poland June 2022
- △ Applied CATS seminar at KTH (virtual) May 2022

Logarithmic Voronoi polytopes for discrete linear models

- † Algebraic Statistics at the University of Hawai'i at Manoa May 2022
- ★ AMS Spring Central Sectional Meeting (virtual) March 2022

Introduction to SAGE

- † Mathematical computing virtual workshop by UCB and UBC Apr 2022

Linear Spaces and Grassmannians

- △ Nonlinear Algebra course at MPI MiS (Leipzig, Germany) June 2019

Caratheodory, Radon, and Helly theorems in convex geometry

- △ Minicourse on Convex Geometry at MPI MiS (Leipzig, Germany) July 2021

	<i>Ice Models for Types A and B</i> (two talks) △ Berkeley Combinatorics Reading Seminar	October 2019
	<i>Combinatorial Nullstellensatz</i> Wesleyan University Thesis Defense	January 2019
	<i>Visibility Graphs of Staircase Polygons</i> † Berkeley Undergraduate Number Theory Conference	April 2018
Research visits	<i>Collaborate@ICERM</i> at Brown University, RI <i>AIM SQuaRE</i> in Santa Clara, CA Institute for Mathematical and Statistical Innovation (IMSI) long-term visitor for <i>Algebraic Statistics and Our Changing World</i> Max Planck Institute for Mathematics in the Sciences	Jul 2026 Aug 2024 Sep–Dec 2023 2019, '20, '22, '24
Skills	Programming Macaulay2, SAGE, Mathematica, Singular, $\LaTeX$, C, C++, Python, Julia, OCaml, SML, HTML Languages English (native), Russian (native), Hebrew (intermediate)	
Service	Organizing <i>Algebraic Geometry: A Window to Machine Learning</i> co-organizer, IPAM Workshop <i>Math machine learning seminar</i> , co-organizer AMS Special Session <i>Algebraic Statistics In Our Changing World</i> AMS Special Session <i>Algebraic Methods in Machine Learning and Optimization</i> co-organizer, <i>Joint Math Meeting</i> (JMM 2025) <i>Berkeley nonlinear algebra seminar</i> , co-organizer Mentorship and outreach Women in math postdoc panel, panelist, <i>UCLA</i> Women in math mentorship program, mentor, <i>UCLA</i> STEMinist club, invited speaker, <i>Berkeley High School</i> Noetherian Ring, member, <i>UC Berkeley</i> Math Club graduate school panel, panelist, <i>Wesleyan University</i> Directed reading program (DRP), mentor, <i>UC Berkeley</i> Unbounded Representation (URP), officer, <i>UC Berkeley</i>	Feb 2027 2024–present Jan 2025 Fall 2022 Mar 2026 2024–2026 Nov 2021 2019–2023 2020 2020 2019–2020

Reviewing

Advances in Applied Mathematics

SIAM Journal on Discrete Mathematics

SIAM Journal on Applied Algebra and Geometry

Algebraic Statistics

Topology Proceedings

The International Conference on Machine Learning (ICML)